Partial Differential Equations

Marc DeCarlo

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Abstract

"Now I am become Death, the destroyer of worlds" - Big O boy

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1 The Heat Equation

The heat equation is a fundamental partial differential equation that describes how heat diffuses through a given region over time. It is given by the following equation:

$$u_t = a^2 u_{xx} \tag{1}$$

This is the one-dimensional heat equation, generally modeling a rod, where $u(x,t)$ represents the temperature at position x and time t, and a^2 is the thermal diffusivity constant. The subscript notation denotes partial derivatives, with u_t representing the partial derivative of u with respect to time, and u_{xx} representing the second partial derivative of u with respect to space. It is usually presented with initial and boundary conditions, which for the following analysis we will use the following:

$$\begin{aligned} u(x,0) &= f(x) \quad \text{for } 0 < x < L \\ u(0,t) &= 0 \quad \text{for } t > 0 \\ u(L,t) &= 0 \quad \text{for } t > 0 \end{aligned}$$

The initial condition $u(x,0) = f(x)$ specifies the temperature distribution along the rod at time $t = 0$, while the boundary conditions $u(0,t) = 0$ and $u(L,t) = 0$ indicate that the ends of the rod are held at zero temperature for all times $t > 0$. The goal is to find a function $u(x,t)$ that satisfies the heat equation (1) along with the given initial and boundary conditions. This problem can be solved using various methods, including separation of variables, Fourier series, and numerical techniques such as finite difference methods.

1.1 Separation of Variables

This is a common technique for solving partial differential equations and the process will be shown for the heat equation (1). The idea is to assume that the solution can be written as a product of two functions, one depending only on x and the other depending only on t. Suppose:

$$u(x,t) = X(x)T(t)$$

where $X(x)$ is a function of x alone and $T(t)$ is a function of t alone. We make no claim about what X and T are yet. Let's transform the heat equation into separation form:

$$\begin{cases} u_t = X(x)T'(t) \\ u_x = X'(x)T(t) \\ u_{xx} = X''(x)T(t) \end{cases} \implies X(x)T'(t) = a^2 X''(x)T(t)$$

This doesn't look like it's helpful yet, but we can rearrange it to get:

$$\frac{T'(t)}{a^2 T(t)} = \frac{X''(x)}{X(x)}$$

so now each side only has one variable. Since our equation must hold for all x and t, we can set both sides equal to a constant, say k:

$$\frac{T'(t)}{a^2 T(t)} = \frac{X''(x)}{X(x)} = k$$

This gives us two ordinary differential equations:

$$\begin{cases} T'(t) - a^2 k T(t) = 0 \\ X''(x) - kX(x) = 0 \end{cases}$$

These are much easier to solve.

1.1.1 What can k be?

The value of the constant k at this point is not known. Let's try to refine what k can be. k can be either of the following: $k > 0$, $k = 0$, or $k < 0$.
Case 1: $k = 0$: In this case, the equation for $X(x)$ becomes:

$$\begin{cases} T'(t) = 0 \\ X''(x) = 0 \end{cases} \implies X(x) = C_1 x + C_2$$

for some constants C_1 and C_2. Let's check the initial and boundary conditions:

$$u(0,t) = X(0)T(t) = C_2 \cdot T(t) = 0$$

For this to hold for all t, we must have $C_2 = 0$. Now we check the other boundary condition:

$$u(L,t) = X(L)T(t) = C_1 L \cdot T(t) = 0$$

For this to hold for all t, we must have $C_1 = 0$. Thus, the only solution for $k = 0$ is the trivial solution $u(x,t) = 0$ which we are not interested in.
Case 2: $k > 0$: From ODEs:

$$X''(x) - kX(x) = 0 \implies X(x) = C_1 e^{\sqrt{k}x} + C_2 e^{-\sqrt{k}x}$$

Check initial and boundary conditions, $u(0,t) = 0$:

$$u(0,t) = X(0)T(t) = (C_1 + C_2)T(t) = 0 \implies C_2 = -C_1 \implies X(x) = C_1(e^{\sqrt{k}x} - e^{-\sqrt{k}x})$$

Check second boundary condition, $u(L,t) = 0$:

$$u(L,t) = X(L)T(t) = C_1(e^{\sqrt{k}L} - e^{-\sqrt{k}L})T(t) = 0$$

Since $e^{\sqrt{k}L} \geq e^{-\sqrt{k}L}$ for $k > 0$, $e^{\sqrt{k}L} - e^{-\sqrt{k}L} \neq 0$, so we must have $C_1 = 0$. Thus, the only solution for $k > 0$ is also the trivial solution $u(x,t) = 0$.
Case 3: $k < 0$: Let $k = -\lambda^2$ for some $\lambda > 0$. Then the equation for $X(x)$ becomes:

$$X''(x) + \lambda^2 X(x) = 0$$

This is a second-order linear homogeneous ODE with constant coefficients. The characteristic equation is:

$$r^2 + \lambda^2 = 0 \implies r = \pm i\lambda$$

Thus, the general solution for $X(x)$ is:

$$X(x) = C_1 \cos(\lambda x) + C_2 \sin(\lambda x)$$

Checking the boundary condition $u(0,t) = 0$:

$$u(0,t) = X(0)T(t) = C_1 T(t) = 0 \implies C_1 = 0 \implies X(x) = C_2 \sin(\lambda x)$$

Now checking the second boundary condition $u(L,t) = 0$:

$$u(L,t) = X(L)T(t) = C_2 \sin(\lambda L)T(t) = 0 \implies \sin(\lambda L) = 0$$

From the unit circle, $\sin(z) = 0$ when $z = n\pi$ for some integer n. Thus, we have:

$$\lambda L = n\pi \implies \lambda = \frac{n\pi}{L}, \quad n \in \mathbb{N}$$

Thus, the values of k that yield non-trivial solutions are:

$$k_n = -\lambda_n^2 = -\left(\frac{n\pi}{L}\right)^2, \quad n \in \mathbb{N}$$

And the solution for $X(x)$ is:

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right), \quad n \in \mathbb{N}$$

These individual X_n are called the eigenfunctions of the problem, and the corresponding k_n are called the eigenvalues.

1.1.2 Finding $T(t)$

$$T' - a^2 kT = 0 \implies T' + a^2\lambda^2 T = 0$$

This is a first-order linear ODE we can solve via separation:

$$\frac{dT}{dt} = -a^2\lambda^2 T \implies \frac{1}{T}\frac{dT}{dt} = -a^2\lambda^2 \implies \int \frac{1}{T} dT = -a^2\lambda^2 \int dt$$

$$\implies \ln|T| = -a^2\lambda^2 t + C \implies T(t) = Ae^{-a^2\lambda^2 t}$$

1.2 Constructing the General Solution

From the previous sections, we have found that the eigenfunctions of the problem are $X_n(x) = \sin\left(\frac{n\pi x}{L}\right)$ and the corresponding time functions are $T_n(t) = e^{-a^2\lambda_n^2 t}$. Thus, the product of these gives us a family of solutions to the heat equation:

$$u_n(x,t) = X_n(x)T_n(t) = \sin\left(\frac{n\pi x}{L}\right) e^{-a^2\left(\frac{n\pi}{L}\right)^2 t}, \quad n \in \mathbb{N}$$

Combining these solutions, we can construct the general solution to the heat equation as an infinite sum of these eigenfunctions:

$$u(x,t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) e^{-a^2\left(\frac{n\pi}{L}\right)^2 t} \tag{2}$$

Checking the final initial condition:

$$u_n(x,0) = \sin\left(\frac{n\pi x}{L}\right) = f(x)$$

depending on the initial condition $f(x)$, this could be just a matter of choosing the right n to match $f(x)$. But what if f is not a trigonometric function? Can we represent an arbitrary f as a sum of trigonometric functions? Enter, **Fourier**! This is how we can find the coefficients b_n in the general solution (2). We do not have the tools yet to do this, but we can after a brief introduction to Fourier series in the next section.

After the next section we will be able to show that the coefficients b_n can be found by the following formula:

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right) dx, \quad n \in \mathbb{N}$$

This now gives us a complete solution to the heat equation (1) with the given initial and boundary conditions.

$$u(x,t) = \sum_{n=1}^{\infty} \left(\frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right) dx\right) \sin\left(\frac{n\pi x}{L}\right) e^{-a^2\left(\frac{n\pi}{L}\right)^2 t} \tag{3}$$

2 Fourier Series

Recall: the general solution to the heat equation (2):

$$u(x,t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) e^{-a^2\left(\frac{n\pi}{L}\right)^2 t}$$

The goal of this section is to build a method for finding the coefficients b_n in the above solution, given an initial condition. To do this, we will need to introduce the concept of Fourier series, which allows us to express a function as an infinite sum of sines and cosines.

2.1 Fourier Series

A Fourier series is a way to represent a periodic function as an infinite sum of sines and cosines. For a function $f(x)$ defined on the interval $[0, L]$, the Fourier series is given by:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)\right)$$

where the coefficients a_n and b_n are given by:

$$a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right) dx, \quad n \geq 0$$

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right) dx, \quad n \geq 1$$

So we want to find the coefficients, b_n, of the Fourier basis functions, $\sin\left(\frac{n\pi x}{L}\right)$, to represent f as a sum of sines.

2.1.1 Why are they a basis?

To show this properly, we need to review some topics from linear algebra through the following definitions.

Definition 2.1. A set of functions $\{f_n\}$ is said to be **linearly independent** if the only way to write the zero function as a linear combination of the f_n is to have all coefficients be zero. In other words, if $\sum_n c_n f_n(x) = 0$ for all x, then $c_n = 0$ for all n.

Definition 2.2. A set of functions $\{f_n\}$ is a **basis** for a function space if every function in the space can be written as a linear combination of the f_n. In other words, for any function g in the space, there exist coefficients c_n such that $g(x) = \sum_n c_n f_n(x)$ for all x.

Definition 2.3. An **inner product** on a function space is the same as an inner product on a vector space, but instead of vectors, we have functions. In this context, the inner product

is often called the **integral inner product**. For two functions f and g, the inner product is defined as:

$$\langle f, g\rangle = \int_0^L f(x)g(x)dx$$

All of the same properties of an inner product hold for this integral inner product, such as linearity, symmetry, and positive-definiteness.

Definition 2.4. Two functions f and g are said to be **orthogonal** with respect to the inner product if $\langle f, g\rangle = 0$. A set of functions $\{f_n\}$ is said to be an **orthogonal set** if every pair of distinct functions in the set is orthogonal. If, in addition, each function in the set has norm 1 (i.e., $\langle f_n, f_n\rangle = 1$), then the set is called an **orthonormal set**.

Lemma 2.5. *An orthogonal set of functions is linearly independent.*

With this dump of definitions and the lemma, we can now say that if we can show that the set of functions $\{\sin\left(\frac{n\pi x}{L}\right) \ : \ n \in \mathbb{N}\}$ is orthogonal with respect to the integral inner product, then we can conclude that they are linearly independent therefore they span some subspace of the function space.

2.1.2 Orthogonality Proof

Lets check orthogonality of the sine functions by choosing two distinct functions from the set, say $\sin\left(\frac{m\pi x}{L}\right)$ and $\sin\left(\frac{n\pi x}{L}\right)$.

$$\langle \sin\left(\frac{m\pi x}{L}\right), \sin\left(\frac{n\pi x}{L}\right)\rangle = \int_0^L \sin\left(\frac{m\pi x}{L}\right)\sin\left(\frac{n\pi x}{L}\right)dx$$

Recall: the trigonometric identity:

$$\sin A \sin B = \frac{1}{2}\left(\cos(A-B) - \cos(A+B)\right)$$

$$\implies \int_0^L \sin\left(\frac{m\pi x}{L}\right)\sin\left(\frac{n\pi x}{L}\right)dx = \frac{1}{2}\int_0^L \left(\cos\left(\frac{(m-n)\pi x}{L}\right) - \cos\left(\frac{(m+n)\pi x}{L}\right)\right)dx$$

This leads us to two cases: $m = n$, and $m \neq n$.
Case 1: $m = n$:

$$\frac{1}{2}\int_0^L \left(\cos(0) - \cos\left(\frac{2n\pi x}{L}\right)\right)dx = \frac{1}{2}\int_0^L \left(1 - \cos\left(\frac{2n\pi x}{L}\right)\right)dx$$

$$= \frac{1}{2}\left[x - \frac{L}{2n\pi}\sin\left(\frac{2n\pi x}{L}\right)\right]_0^L = \frac{L}{2}$$

which is not zero, so the functions are not orthogonal when $m = n$.
Case 2: $m \neq n$:

$$\frac{1}{2}\int_0^L \left(\cos\left(\frac{(m-n)\pi x}{L}\right) - \cos\left(\frac{(m+n)\pi x}{L}\right)\right)dx$$

$$= \frac{1}{2}\left[\frac{L}{(m-n)\pi}\sin\left(\frac{(m-n)\pi x}{L}\right) - \frac{L}{(m+n)\pi}\sin\left(\frac{(m+n)\pi x}{L}\right)\right]_0^L$$

Since $(m+n)$ and $(m-n)$ are both integers, the sine terms will evaluate to zero at both limits of integration, so the entire expression evaluates to zero. Thus,

$$\langle \sin\left(\frac{m\pi x}{L}\right), \sin\left(\frac{n\pi x}{L}\right)\rangle = 0 \quad \text{for } m \neq n$$

2.2 Finding the Coefficients

Now that we have shown that the sine functions are orthogonal, we can use this property to find the coefficients b_n in the Fourier series expansion of $f(x)$. We want to find b_n such that:

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

To find b_n, we can take the inner product of both sides of the equation with $\sin\left(\frac{m\pi x}{L}\right)$ for some integer m:

$$\langle f, \sin\left(\frac{m\pi x}{L}\right)\rangle = \langle \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right), \sin\left(\frac{m\pi x}{L}\right)\rangle$$

Using linearity of the inner product, we can pull the sum and the coefficients out:

$$\langle f, \sin\left(\frac{m\pi x}{L}\right)\rangle = \sum_{n=1}^{\infty} b_n \langle \sin\left(\frac{n\pi x}{L}\right), \sin\left(\frac{m\pi x}{L}\right)\rangle$$

From the orthogonality proof, we know that $\langle \sin\left(\frac{n\pi x}{L}\right), \sin\left(\frac{m\pi x}{L}\right)\rangle = 0$ for $n \neq m$ and $\langle \sin\left(\frac{n\pi x}{L}\right), \sin\left(\frac{n\pi x}{L}\right)\rangle = \frac{L}{2}$ for $n = m$. Thus, the only term in the sum that survives is the one where $n = m$, giving us:

$$\langle f, \sin\left(\frac{m\pi x}{L}\right)\rangle = b_m \langle \sin\left(\frac{m\pi x}{L}\right), \sin\left(\frac{m\pi x}{L}\right)\rangle = b_m \frac{L}{2}$$

Solving for b_m, we get:

$$b_m = \frac{2}{L}\langle f, \sin\left(\frac{m\pi x}{L}\right)\rangle = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{m\pi x}{L}\right) dx$$

Thus, we have found the formula for the coefficients b_n in the Fourier series expansion of $f(x)$:

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right) dx, \quad n \in \mathbb{N}$$